n-channel JFETs *designed for . . .*

Siliconix

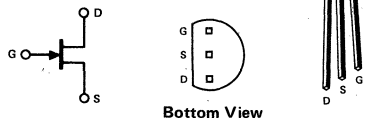
Performance Curves NRL, NPA See Section 4

- **General Purpose Amplifiers**
- **Switches**

BENEFITS
- Low Cost
- Automated Insertion Package

TO-92
See Section 6

Plastic



Bottom View

*ABSOLUTE MAXIMUM RATINGS (25°C)

Drain-Source Voltage	25 V
Drain-Gate Voltage	25 V
Source-Gate Voltage	25 V
Total Device Dissipation at 25°C	310 mW
Derate above 25°C	2.82 mW/°C
Operating Junction Temperature	135°C
Storage Temperature Range	-65 to +150°C

*ELECTRICAL CHARACTERISTICS (25°C unless otherwise noted)

		Characteristic		2N5457			2N5458			2N5459			Unit	Test Conditions	
				Min	Typ	Max	Min	Typ	Max	Min	Typ	Max			
1	STATIC	I_{GSS}	Gate Reverse Current		-0.01	-1.0		-0.01	-1.0		-0.01	-1.0	nA	V_{GS} = -15 V, V_{DS} = 0	
2						-200			-200			-200			T_A = +100°C
3		BV_{GSS}	Gate-Source Breakdown Voltage	-25	-60		-25	-60		-25	-60		V	I_G = -10 μA, V_{DS} = 0	
4		$V_{GS(off)}$	Gate-Source Cutoff Voltage	-0.5		-6.0	-1.0		-7.0	-2.0		-8.0		V_{DS} = 15 V, I_D = 10 nA	
5		I_{DSS}	Saturation Drain Current	1.0		5.0	2.0		9.0	4.0		16	mA	V_{DS} = 15 V, V_{GS} = 0 (Note 1)	
6	DYNAMIC	g_{fs}	Common-Source Forward Transconductance	1,000		5,000	1,500		5,500	2,000		6,000	μmho	V_{DS} = 15 V, V_{GS} = 0	f = 1 kHz
7		g_{os}	Common-Source Output Conductance		10	50		15	50		20	50			
8		C_{iss}	Common-Source Input Capacitance		4.5	7.0		4.5	7.0		4.5	7.0	pF		f = 1 MHz
9		C_{rss}	Common-Source Reverse Transfer Capacitance		1.0	3.0		1.0	3.0		1.0	3.0			
10		NF	Noise Figure		0.04	3.0		0.04	3.0		0.04	3.0	dB	V_{DS} = 15 V, V_{GS} = 0 R_G = 1 MΩ, NBW = 1 Hz	f = 1 kHz

*JEDEC registered data

NOTE:

1. Pulse test pulsewidth = 2 ms.

NRL, NPA

2

n-channel JFETs *designed for . . .*

- **VHF/UHF Amplifiers**
- **Mixers**
- **Oscillators**
- **Analog Switches**

Siliconix

Performance Curves NH See Section 4

BENEFITS

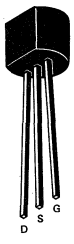
- Low Cost
- Completely Specified for 400 MHz Operation
- Low Error Analog Switch
  Very Little Charge Coupling
  $C_{rss} < 1.0$ pF

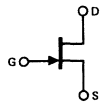
***ABSOLUTE MAXIMUM RATINGS (25°C)**

Drain-Gate Voltage . . . . . 25 V
Source Gate Voltage . . . . . 25 V
Drain Current . . . . . 30 mA
Forward Gate Current . . . . . 10 mA
Total Device Dissipation @ 25°C . . . . . 360 mW
Derate above 25°C . . . . . 3.27 mW/°C
Operating Junction Temperature Range . . . . . −65 to +135°C
Storage Temperature Range . . . . . −65 to +150°C
Lead Temperature
(1/16" from case for 10 seconds) . . . . . 240°C

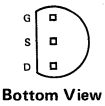
Plastic

TO-92
See Section 6





Bottom View



***ELECTRICAL CHARACTERISTICS (25°C unless otherwise noted)**

		Characteristic		2N5484 Min	2N5484 Max	2N5485 Min	2N5485 Max	2N5486 Min	2N5486 Max	Unit	Test Conditions	
1	STATIC	I_{GSS}	Gate Reverse Current		-1.0		-1.0		-1.0	nA	V_{GS} = -20 V, V_{DS} = 0	
2					-200		-200		-200			T_A = +100°C
3		BV_{GSS}	Gate-Source Breakdown Voltage	-25		-25		-25		V	I_G = -1 μA, V_{DS} = 0	
4		$V_{GS(off)}$	Gate-Source Cutoff Voltage	-0.3	-3.0	-0.5	-4.0	-2.0	-6.0		V_{DS} = 15 V, I_D = 10 nA	
5		I_{DSS}	Saturation Drain Current	1.0	5.0	4.0	10	8.0	20	mA	V_{DS} = 15 V, V_{GS} = 0 (Note 1)	
6	DYNAMIC	g_{fs}	Common-Source Forward Transconductance	3,000	6,000	3,500	7,000	4,000	8,000	μmhos	V_{DS} = 15 V, V_{GS} = 0	f = 1 kHz
7		g_{os}	Common-Source Output Conductance		50		60		75			
8		$Re(y_{fs})$	Common-Source Forward Transconductance	2,500								f = 100 MHz
9						3,000		3,500				f = 400 MHz
10		$Re(y_{os})$	Common-Source Output Conductance		75							f = 100 MHz
11							100		100			f =400 MHz
12		$Re(y_{is})$	Common-Source Input Conductance		100							f = 100 MHz
13							1,000		1,000			f = 400 MHz
14		C_{iss}	Common-Source Input Capacitance		5.0		5.0		5.0	pF		f = 1 MHz
15		C_{rss}	Common-Source Reverse Transfer Capacitance		1.0		1.0		1.0			
16		C_{oss}	Common-Source Output Capacitance		2.0		2.0		2.0			
17		NF	Noise Figure		2.5		2.5		2.5	dB	V_{DS} = 15 V, V_{GS} = 0, R_G = 1 MΩ	f = 1 kHz
18					3.0						V_{DS} = 15 V, I_D = 1 mA, R_G = 1 kΩ	f = 100 MHz
19							2.0		2.0		V_{DS} = 15 V, I_D = 4 mA, R_G = 1 kΩ	
20							4.0		4.0			f = 400 MHz
21		G_{ps}	Common-Source Power Gain	16	25						V_{DS} = 15 V, I_D = 1 mA	f = 100 MHz
22						18	30	18	30		V_{DS} = 15 V, I_D = 4 mA	
23						10	20	10	20			f = 400 MHz

* JEDEC registered data

NH

NOTE:

1 Pulse Test PW 300 μs, duty cycle ≤ 3%

n-channel JFETs *designed for . . .*

■ General Purpose Amplifiers

Siliconix

Performance Curves NPA See Section 4

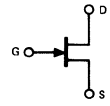
BENEFITS

- High Input Impedance
  I_G = 35 pA Typical
- Good for Low Power Supply Operation
  $V_{GS(off)} < 1.5$ V (J201)

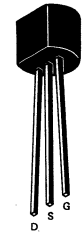
ABSOLUTE MAXIMUM RATINGS (25°C)

Gate-Drain or Gate-Source Voltage (Note 1) . . . . . -40 V
Gate Current . . . . . 50 mA
Total Device Dissipation at 25°C Ambient
(Derate 3.27 mW/°C). . . . . 360 mW
Operating Temperature Range. . . . . –55 to 135°C
Storage Temperature Range. . . . . –55 to 150°C
Lead Temperature Range
(1/16" from case for 10 seconds) . . . . . 300°C

TO-92
See Section 6



Plastic





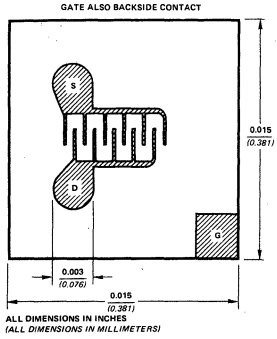
Bottom View

ELECTRICAL CHARACTERISTICS (25°C unless otherwise noted)

		Characteristic		J201			J202			J203			J204			Unit	Test Conditions	
				Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max			
1	STATIC	I_{GSS}	Gate Reverse Current (Note 2)			-100			-100			-100			-100	pA	V_{DS} = 0, V_{GS} = -20 V	
2	STATIC	$V_{GS(off)}$	Gate-Source Cutoff Voltage	-0.3		-1.5	-0.8		-4.0	-2.0		-10.0	-0.3		-2.0	V	V_{DS} = 20 V, I_D = 10 nA	
3	STATIC	BV_{GSS}	Gate-Source Breakdown Voltage	-40			-40			-40			-25			V	V_{DS} = 0, I_G = -1 μA	
4	STATIC	I_{DSS}	Saturation Drain Current (Note 3)	0.2		1.0	0.9		4.5	4.0		20	0.2	1.2	3	mA	V_{DS} = 20 V, V_{GS} = 0	
5	STATIC	I_G	Gate Current (Note 2)		-35			-35			-35			-35		pA	V_{DG} = 20 V, I_D = $I_{DSS(min)}$	
6	DYNAMIC	g_{fs}	Common-Source Forward Transconductance (Note 3)	500			1,000			1,500			500	1500		μmho	V_{DS} = 20 V, V_{GS} = 0	f = 1 kHz
7	DYNAMIC	g_{os}	Common-Source Output Conductance		1			3.5			10			2.5		μmho	V_{DS} = 20 V, V_{GS} = 0	f = 1 kHz
8	DYNAMIC	C_{iss}	Common-Source Input Capacitance		4			4			4			4		pF	V_{DS} = 20 V, V_{GS} = 0	f = 1 MHz
9	DYNAMIC	C_{rss}	Common-Source Reverse Transfer Capacitance		1			1			1			1		pF	V_{DS} = 20 V, V_{GS} = 0	f = 1 MHz
10	DYNAMIC	$\overline{e_n}$	Equivalent Short-Circuit Input Noise Voltage		5			5			5			10		$\frac{nV}{\sqrt{Hz}}$	V_{DS} = 10 V, V_{GS} = 0	f = 1 kHz

NOTES:
1. Geometry is symmetrical. Units may be operated with source and drain leads interchanged.
2. Approximately doubles for every 10°C increase in T_A.
3. Pulse test duration = 2 ms.

NPA



n-channel JFET
designed for . . .

- VHF/UHF Amplifiers
- Oscillators
- Mixers
- Low Input Capacitance High Speed Switch

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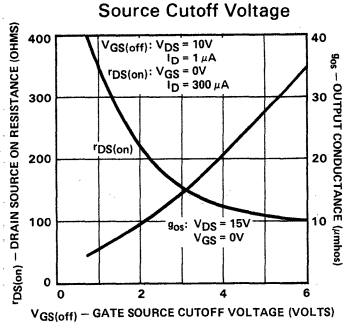
BENEFITS:

- **Low Noise**
  NF = 3 dB Typical @ 400 MHz
- **Wideband**
  High g_{fs}/C_{iss} Ratio

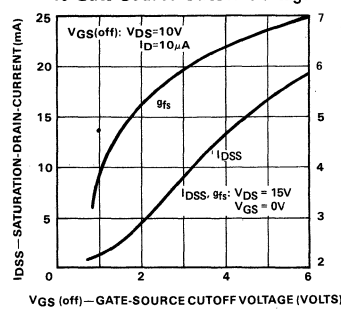
TYPE	PACKAGE	PRINCIPAL DEVICES
Single	TO-72	2N3966, 2N4416-16A 2N3819
Single	TO-92	2N5484-6, 2N5555, 2N5668-70, MPF102, MPF108, MPF112, PN4416, J304-5,
Single	Chip	All of the above devices

PERFORMANCE CURVES (25°C unless otherwise noted)

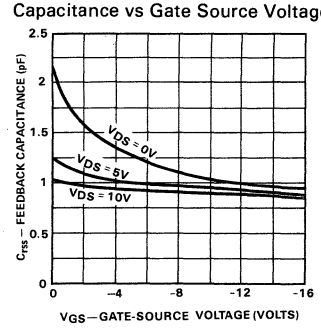
On Resistance & Output Conductance vs Gate-Source Cutoff Voltage



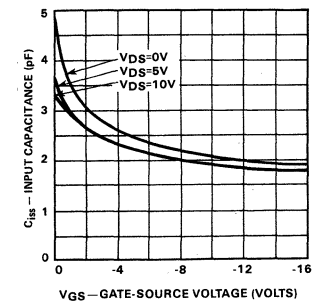
Drain Current & Transconductance vs Gate Source Cutoff Voltage



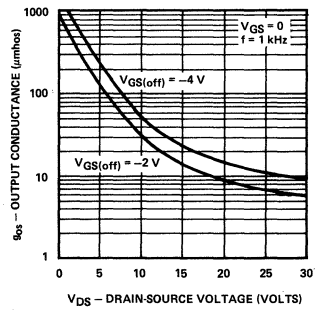
Common Source Reverse Feedback Capacitance vs Gate Source Voltage



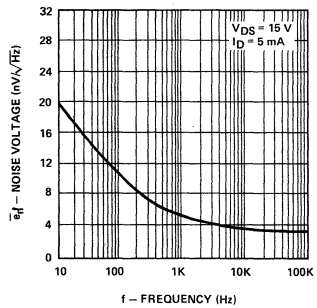
Common Source Input Capacitance vs Gate-Source Voltage



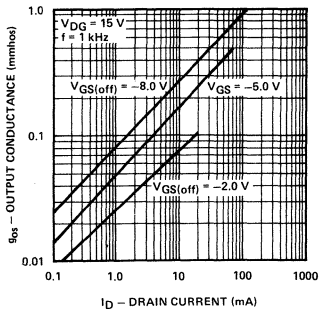
Common-Source Output Conductance vs Drain-Source Voltage



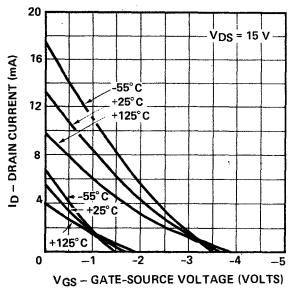
Equivalent Input Noise Voltage vs Frequency



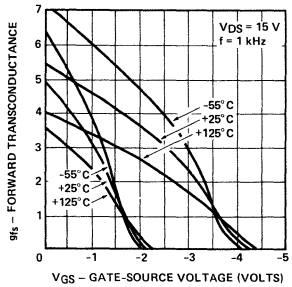
Common-Source Output Conductance vs Drain Current

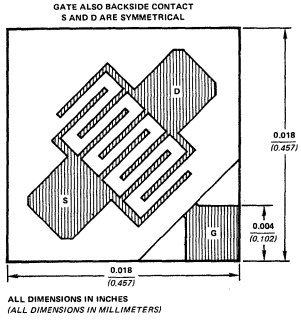


Transfer Characteristics



Transconductance Characteristics





n-channel JFET
designed for . . .

- Small Signal Amplifiers
- VHF Amplifiers
- Oscillators
- Mixers
- Switches

Siliconix

BENEFITS:

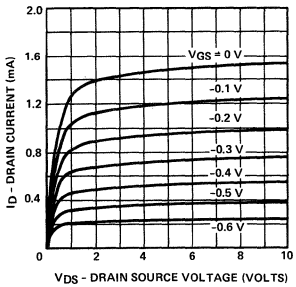
- Wide Input Dynamic Range
  High I_G Breakpoint Voltage
- High Gain
- Low Insertion Loss Switches

TYPE	PACKAGE
Single	TO-72
Single	TO-92
Single	Chip

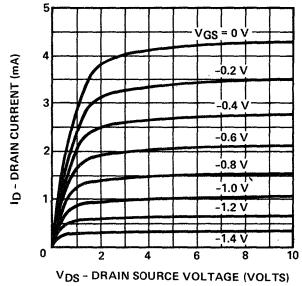
PRINCIPAL DEVICES
2N3821-4, 2N3921-22, 2N4220-2
2N4220A-2A, 2N4223-24
2N3819, 2N5457-9
MPF109, MPF111
All of the above

PERFORMANCE CURVES (25°C unless otherwise noted)

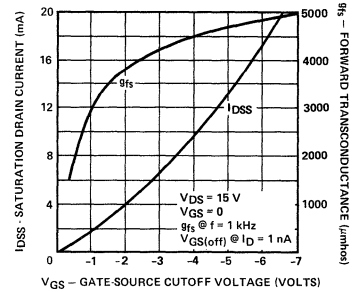
Output Characteristic
($V_{GS(off)}$ = −0.8V)



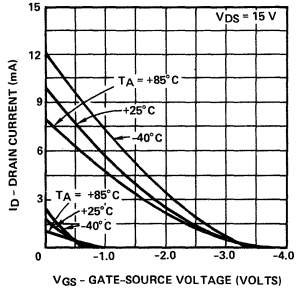
Output Characteristic
($V_{GS(off)}$ = −2.0V)



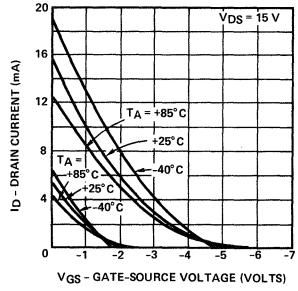
Drain Current & Transconductance vs
Gate-Source Cutoff Voltage



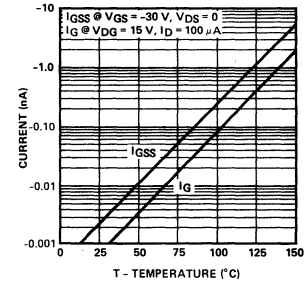
Transfer Characteristic



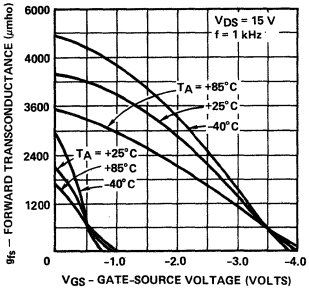
Transfer Characteristics



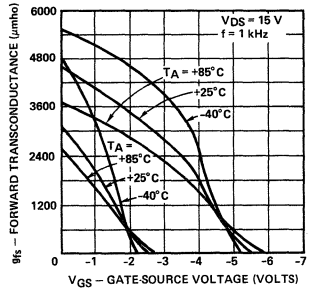
Leakage Currents vs
Ambient Temperature



Transconductance Characteristics



Transconductance Characteristics



'ON' Resistance vs
Ambient Temperature

